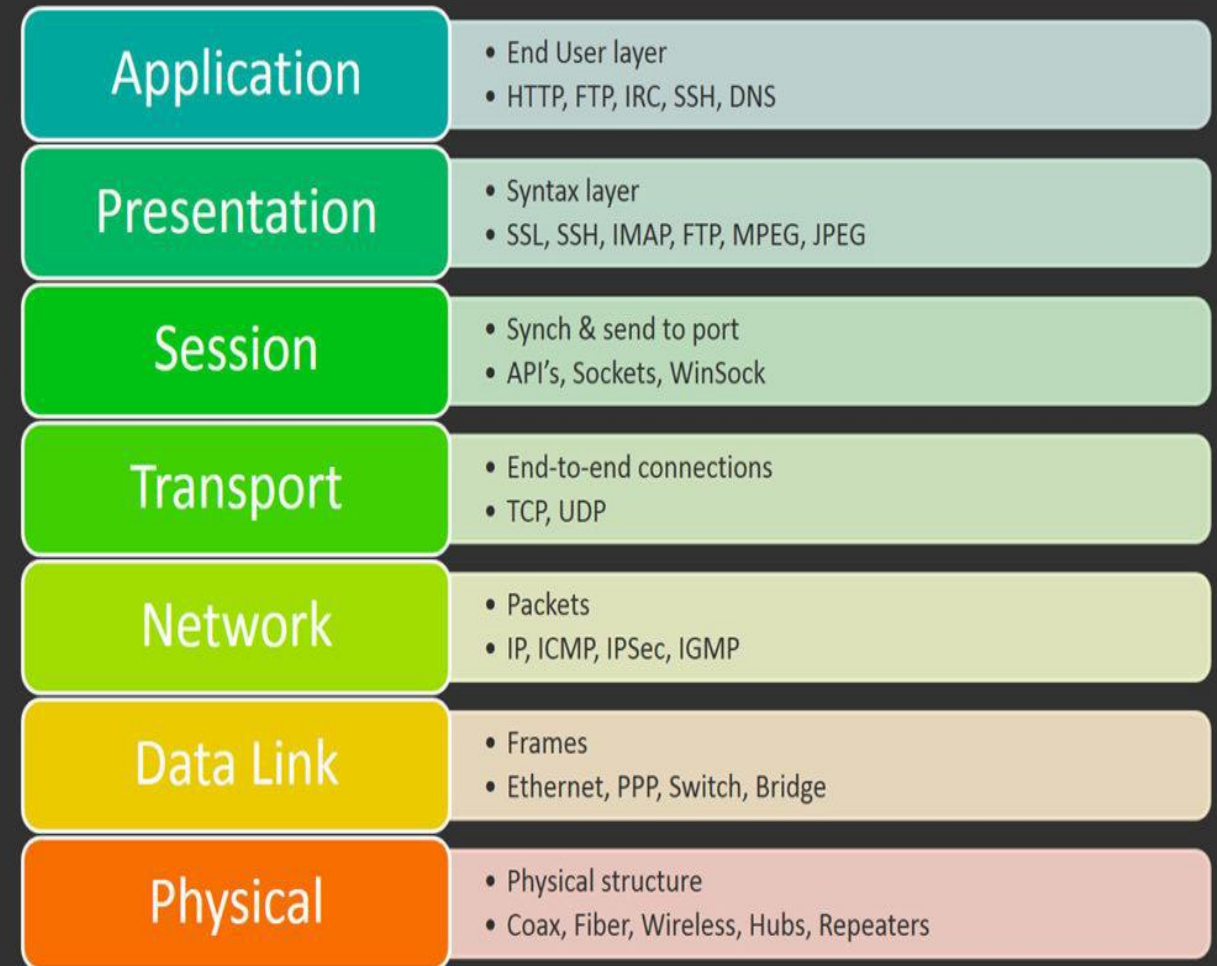


Introduction To Networking

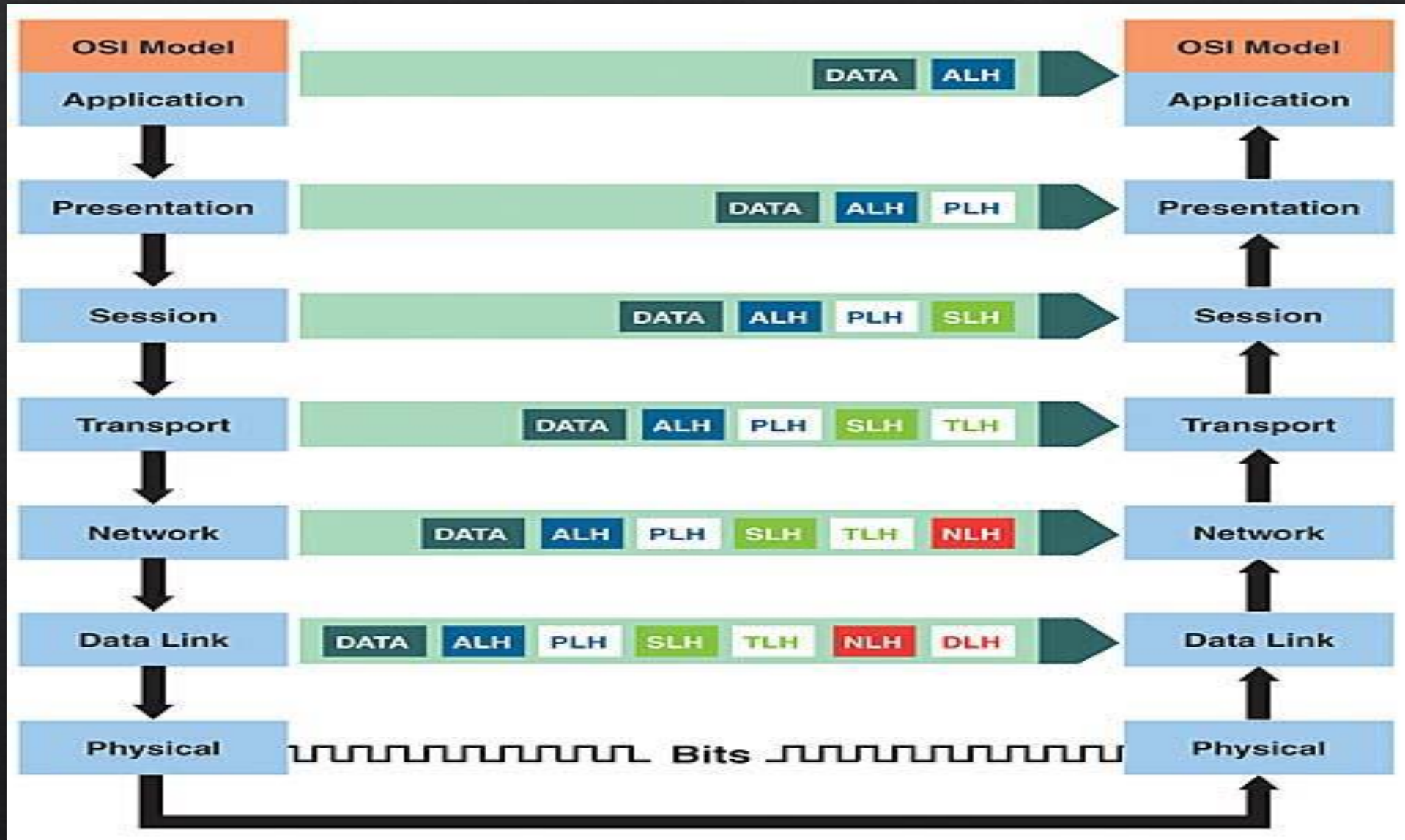
OSI Model

- ◇ What is network?
 - ◇ Network is a way to communicate between systems.
- ◇ What is OSI model?
 - ◇ OSI model is how systems communicate between each others.

7 Layers of the OSI Model



How systems communicate through this layers?



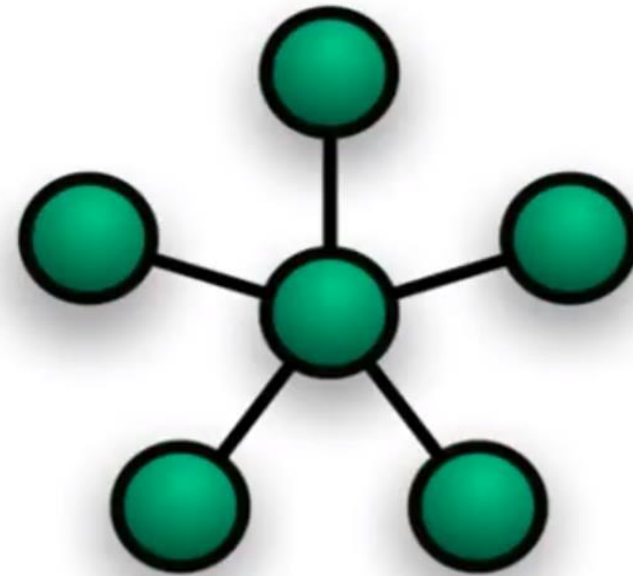
Physical Layer (1)

- ◆ Physical layer is how we physically connect devices (Ethernet cables, Wireless).
- ◆ Ethernet cards and Wi-Fi and network hubs all operate at the physical layer.
- ◆ There are multiple network topologies that we can use to connect devices physically:
 - ◆ Star topology.
 - ◆ Ring topology.
 - ◆ Bus topology.

Network Topologies

Star Topology

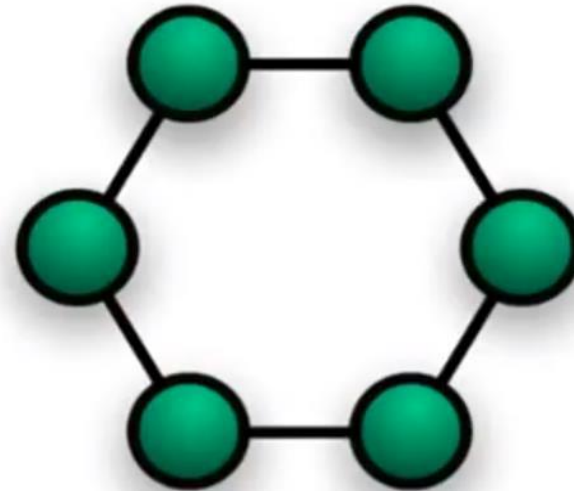
- Each node is connected to a central node (such as a hub or switch)
- Star Topologies provide better performance, isolation of devices, and easy expansion
- However, they have a single point of failure



Network Topologies

Ring Topology

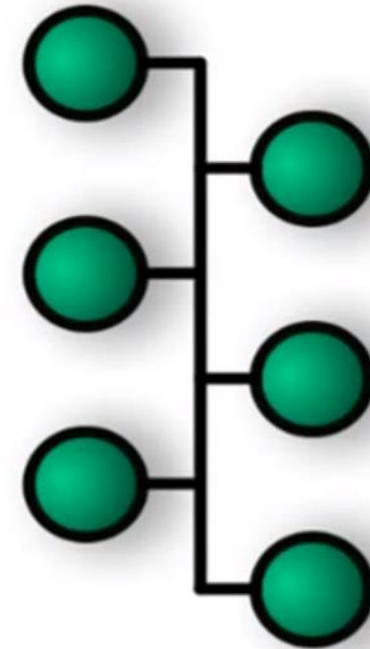
- Each node is connected to two other nodes
- Data travels in one direction, passing through each node to reach its destination
- No central node
- If one node breaks, it can disrupt the entire network



Network Topologies

Bus Topology

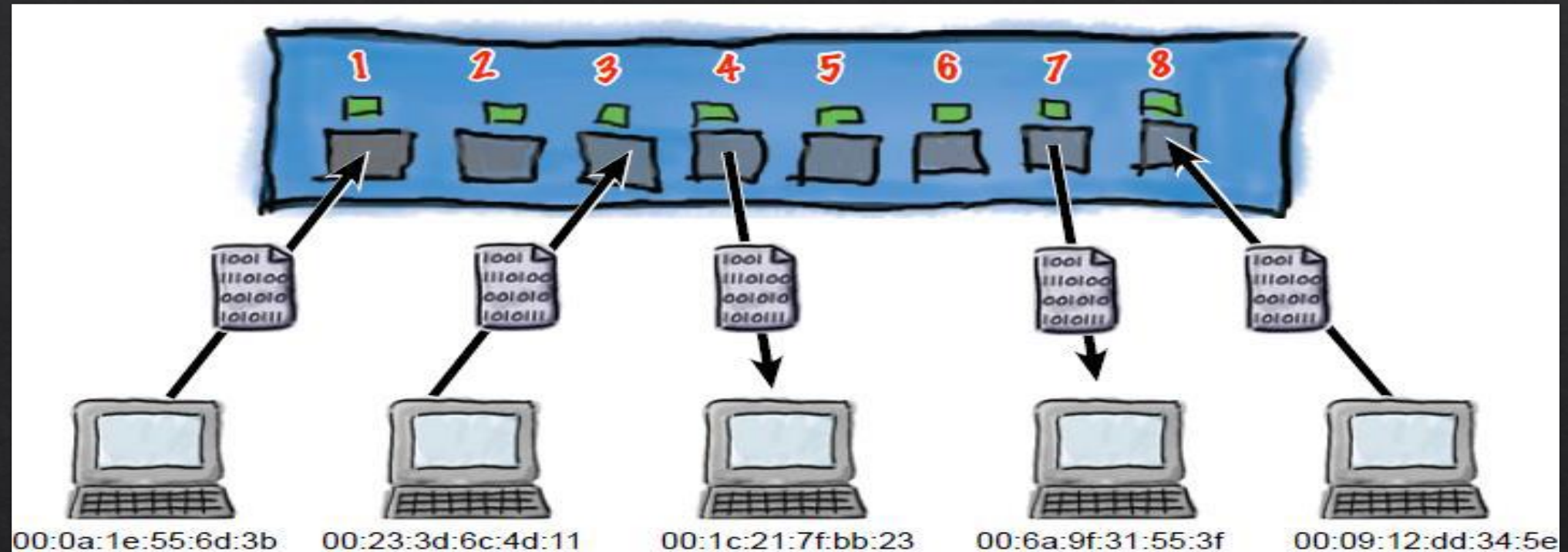
- Each node is connected to a single cable, which all nodes share
- The signal travels in both directions to all nodes
- Only one node can transmit at a time
- Since all nodes share the same bandwidth, performance degrades as more nodes are added
- The bus is a single point of failure



Data Link Layer (2)

- ◆ It responsible to make devices communicate with each others through the same network but not outside the network.
- ◆ Each device has its own MAC address (00:11:22:33:44:55) and through this MAC the devices can communicate and send messages to each other.
- ◆ The first 3 bytes are the manufacturer and the other 3 bytes are the device ID.
- ◆ MAC address can be spoofed.
- ◆ The data in the network layer 2 (Data Link) called Frames.
- ◆ Network switches and Access Points operate at layer 2.

Switches

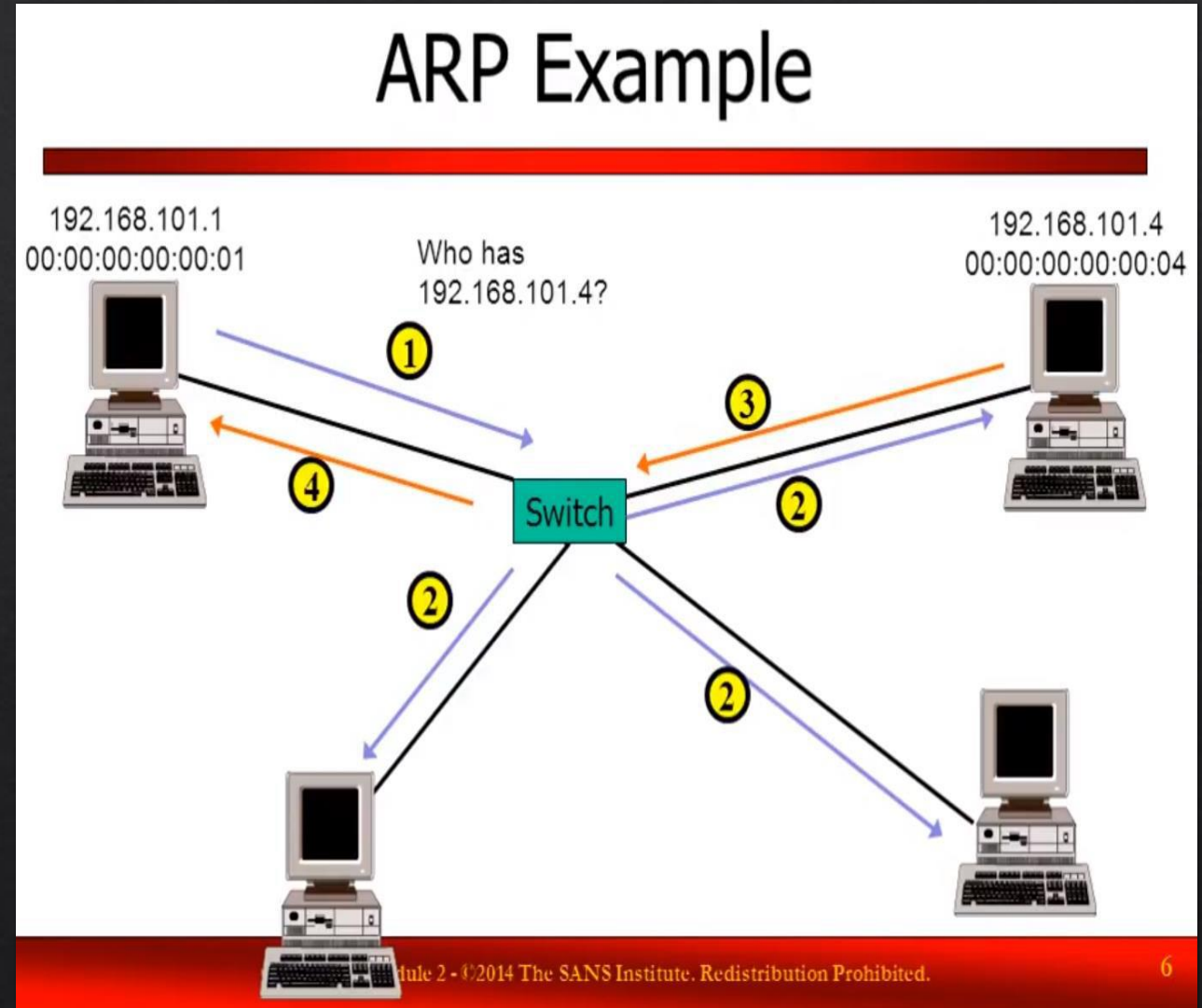


CAM Table

MAC address	Port
00:0a:1e:55:6d:3b	1
00:23:3d:6c:4d:11	3
00:1c:21:7f:bb:23	4
00:6a:9f:31:55:3f	7
00:09:12:dd:34:5e	8

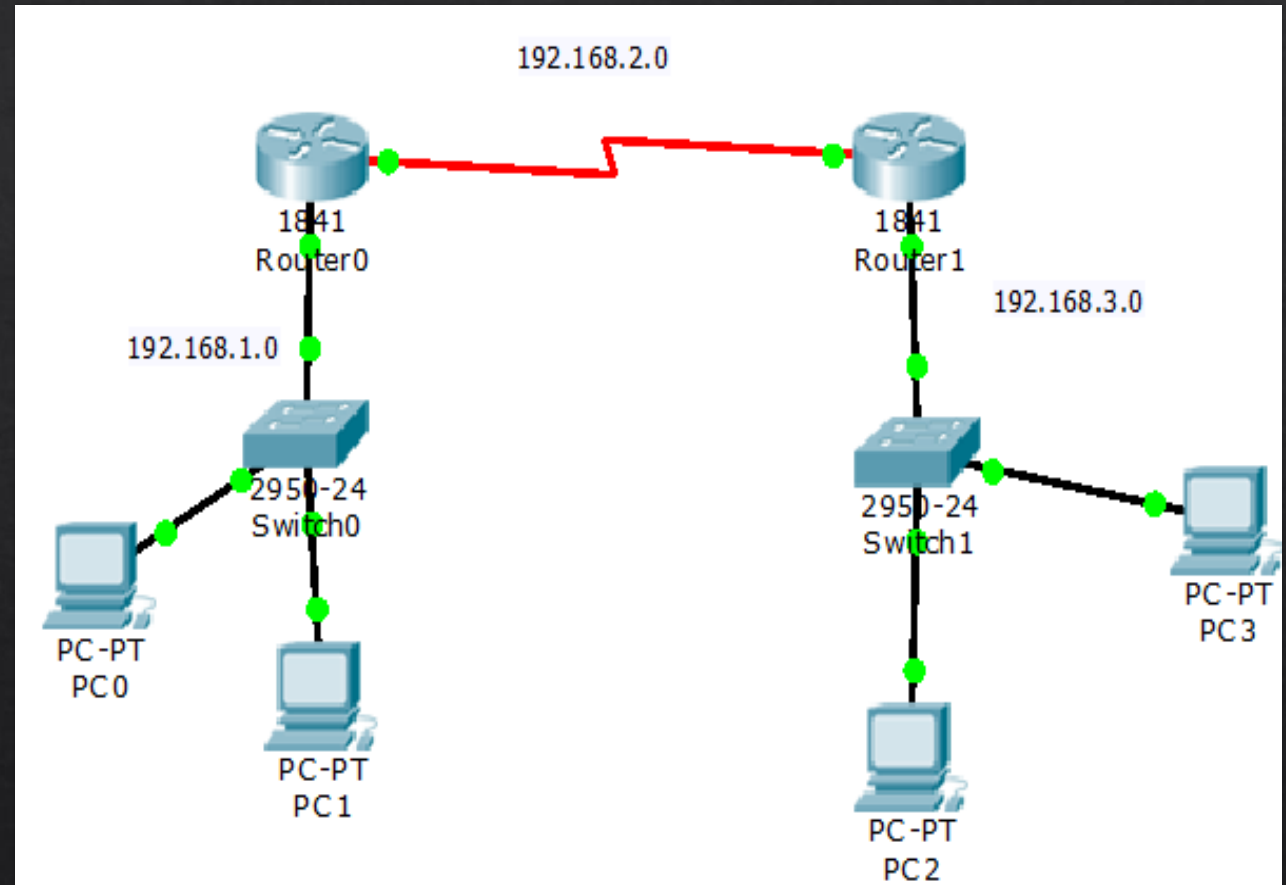
ARP Protocol

- ◆ Find MAC address of the IP address.
- ◆ One system broadcast an ARP request to all the systems in the network asking who has this IP address (192.168.1.1), the system who has this IP respond with its MAC address, then both the systems store each other's MAC address in their ARP cache so they don't have to ask again for a while.
- ◆ The problem with ARP protocol that it accepts responses without even it sends requests (ARP Spoofing).



Network Layer (3)

- ◈ Connect separate networks together.
- ◈ Routers operate at this layer.
- ◈ Devices at this layer communicate to each other through the IP addresses (192.168.1.1).



IPv4 Protocol

◈ 0-255 . 0-255 . 0-255 . 0-255

◈ 8bit . 8bit . 8bit . 8bit

the number

255

in binary is



IP Classes

Class	Range	Example	Notes
A	0-127	0.0.0.0 – 127.0.0.0	First 8bit is the network (N.0.0.0)
B	128-191	128.0.0.0 – 191.255.0.0	First 16bit is the network (N.N.0.0)
C	192-223	192.0.0.0 – 223.255.255.0	First 24bit is the network (N.N.N.0)
D	224-239	224.0.0.0 – 239.255.255.255	Used for multicast.
E	240-255	240.0.0.0 – 255.255.255.255	Reserved for experimental.

Reserved Addresses

- ◇ 127.0.0.0 - 127.255.255.255
 - ◇ Reserved for localhost.
- ◇ 10.0.0.0 - 10.255.255.255
 - ◇ Reserved for local network (LAN).
- ◇ 172.16.0.0 - 172.31.255.255
 - ◇ Reserved for local network (LAN).
- ◇ 192.168.0.0 - 192.168.255.255
 - ◇ Reserved for local network (LAN).

- ◇ First IP of any network is reserved for the network itself (Network Identifier).
- ◇ Last IP of any network is reserved for the broadcast (Broadcast Address).

Class	Network ID	Broadcast Address
A	10.0.0.0	10.255.255.255
B	172.16.0.0	172.16.255.255
C	192.168.1.0	192.168.1.255

Subnet Mask

- It identifies which part of the IP address is the network identifier and which part is the host identifier.
- In normal case would be the same as the network identifier.
- Net mask:
 - Class A 255.0.0.0
 - Class B 255.255.0.0
 - Class C 255.255.255.0

SUBNET MASK IN IP ADDRESSING

Class A	Network	Host	Host	Host
Subnet Mask	255	0	0	0

Class B	Network	Network	Host	Host
Subnet Mask	255	255	0	0

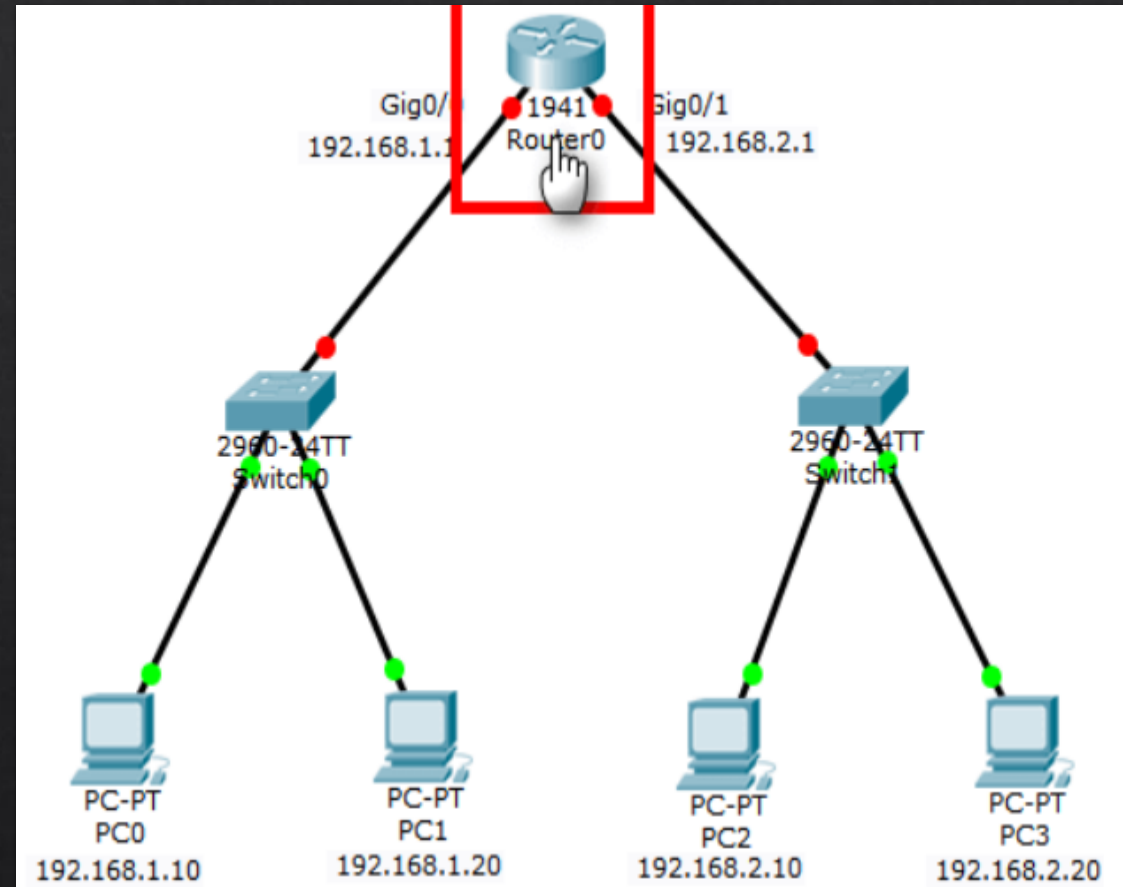
Class C	Network	Network	Network	Host
Subnet Mask	255	255	255	0

CIDR

- ◇ Classless Inter-Domain Routing (CIDR).
- ◇ Another and simple way to represent the subnet mask.
- ◇ Class A:
 - ◇ 10.0.0.0/8
 - ◇ “/8” is the number of bits reserved for the network == 255.0.0.0
- ◇ Class B:
 - ◇ 172.16.0.0/16
 - ◇ “/16” is the number of bits reserved for the network == 255.255.0.0
- ◇ Class C:
 - ◇ 192.168.1.0/24
 - ◇ “/24” is the number of bits reserved for the network == 255.255.255.0

Default Gateway

- ◇ It's responsible for routing you from one network to another network.
- ◇ In most cases it will be the first IP in the network:
 - ◇ Class C: 192.168.1.1
 - ◇ Class B: 172.16.0.1
 - ◇ Class A: 10.0.0.1



DHCP

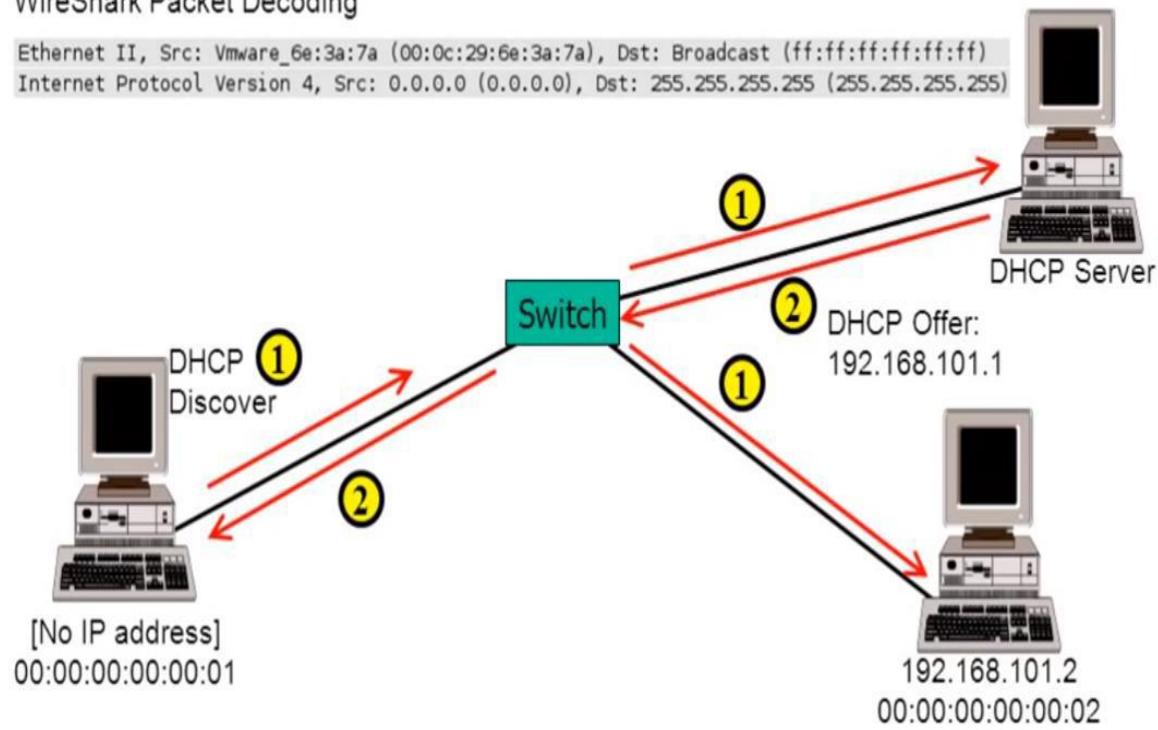
- ◆ DHCP (Dynamic Host Configuration Protocol).
- ◆ Automatically assign to you:
 - ◆ IP address: 192.168.1.10
 - ◆ Subnet mask: 255.255.255.0
 - ◆ default gateway: 192.168.1.1
 - ◆ DNS server: 8.8.8.8, 8.8.4.4
- ◆ In your local network the DHCP server will be in the router itself, but in other enterprise networks it will be a separate server.

How DHCP Work?

DHCP Example (1)

WireShark Packet Decoding

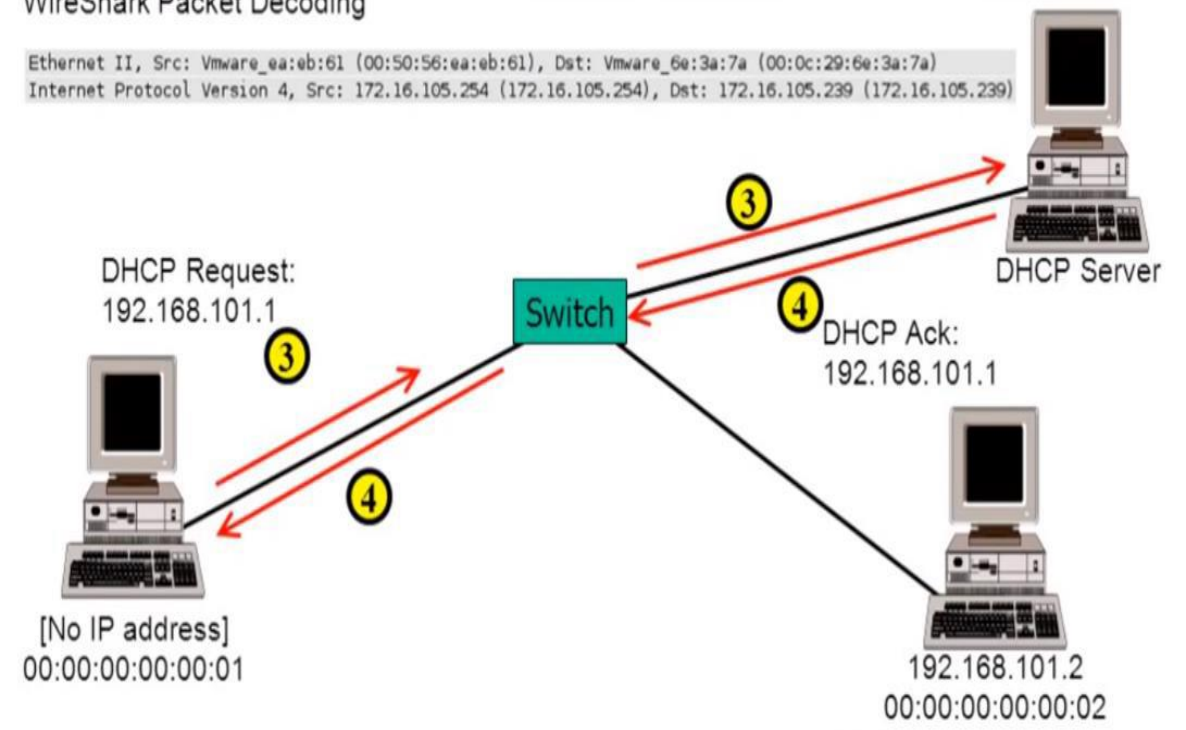
```
Ethernet II, Src: Vmware_6e:3a:7a (00:0c:29:6e:3a:7a), Dst: Broadcast (ff:ff:ff:ff:ff:ff)  
Internet Protocol Version 4, Src: 0.0.0.0 (0.0.0.0), Dst: 255.255.255.255 (255.255.255.255)
```



DHCP Example (2)

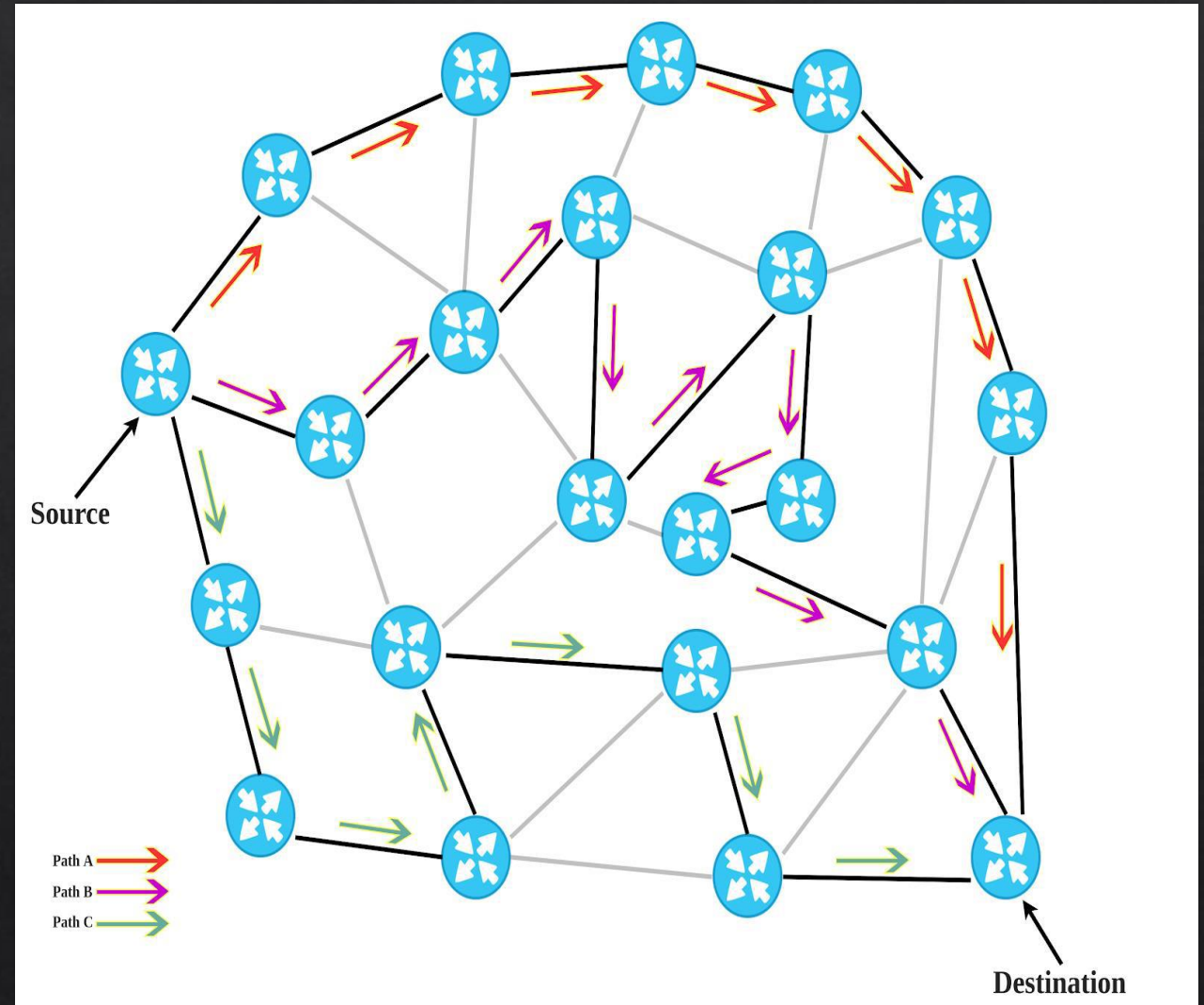
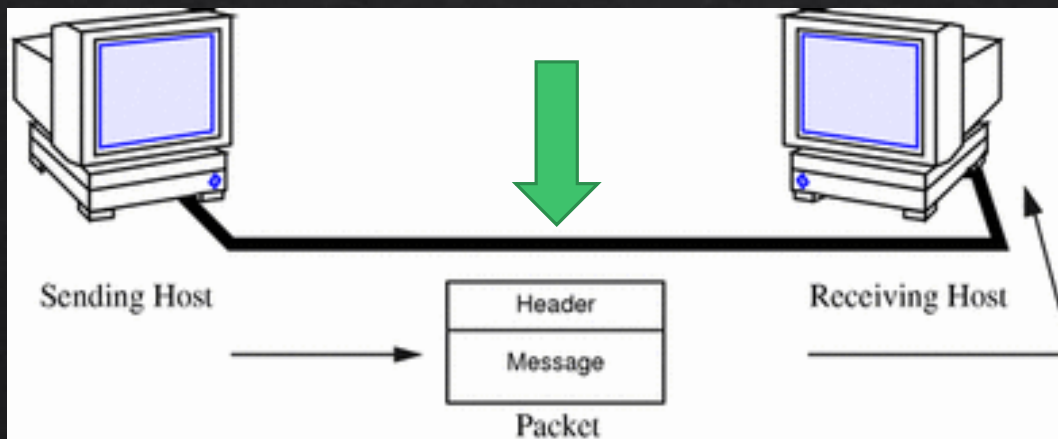
WireShark Packet Decoding

```
Ethernet II, Src: Vmware_ea:eb:61 (00:50:56:ea:eb:61), Dst: Vmware_6e:3a:7a (00:0c:29:6e:3a:7a)  
Internet Protocol Version 4, Src: 172.16.105.254 (172.16.105.254), Dst: 172.16.105.239 (172.16.105.239)
```



Routing

- ◇ Routing is the process of moving packets between networks.
- ◇ Router is the device that routes the packets between networks.
- ◇ Router have multiple interfaces and can connect multiple networks at the same time.



TTL

- ◆ TTL (Time to Live)
- ◆ TTL has fixed number almost 64.
- ◆ It started at 64 then decrement by 1 at every router (hop or node) until it reach its destination.
- ◆ this number exist to prevent the infinite loops.

```
C:\Users\hassan>ping google.com

Pinging google.com [172.217.21.78] with 32 bytes of data:
Reply from 172.217.21.78: bytes=32 time=544ms TTL=53
Reply from 172.217.21.78: bytes=32 time=516ms TTL=53
Reply from 172.217.21.78: bytes=32 time=646ms TTL=53
Reply from 172.217.21.78: bytes=32 time=538ms TTL=53

Ping statistics for 172.217.21.78:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 516ms, Maximum = 646ms, Average = 561ms

C:\Users\hassan>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:
Reply from 192.168.1.1: bytes=32 time=1ms TTL=64
Reply from 192.168.1.1: bytes=32 time=2ms TTL=64
Reply from 192.168.1.1: bytes=32 time=1ms TTL=64
Reply from 192.168.1.1: bytes=32 time=2ms TTL=64
```

Routing Table

- ◇ It's a table in the router device like the CAM table in the switch device.
- ◇ Which is responsible to determine the next hop (Router) that it should send to it the packet (message) to reach the final destination.
- ◇ If no route to the destination network is exist, the router will send the packet to its default gateway.
- ◇ But what will happen if there is more than one route to the destination address?
 - ◇ The router will choose the shortest route.

Where to route to 192.168.1.20 ???

192.168.1.20/32 *via* 10.254.2.1



192.168.1.0/24 *via* 10.254.3.1

192.168.1.30/32 *via* 10.254.3.1

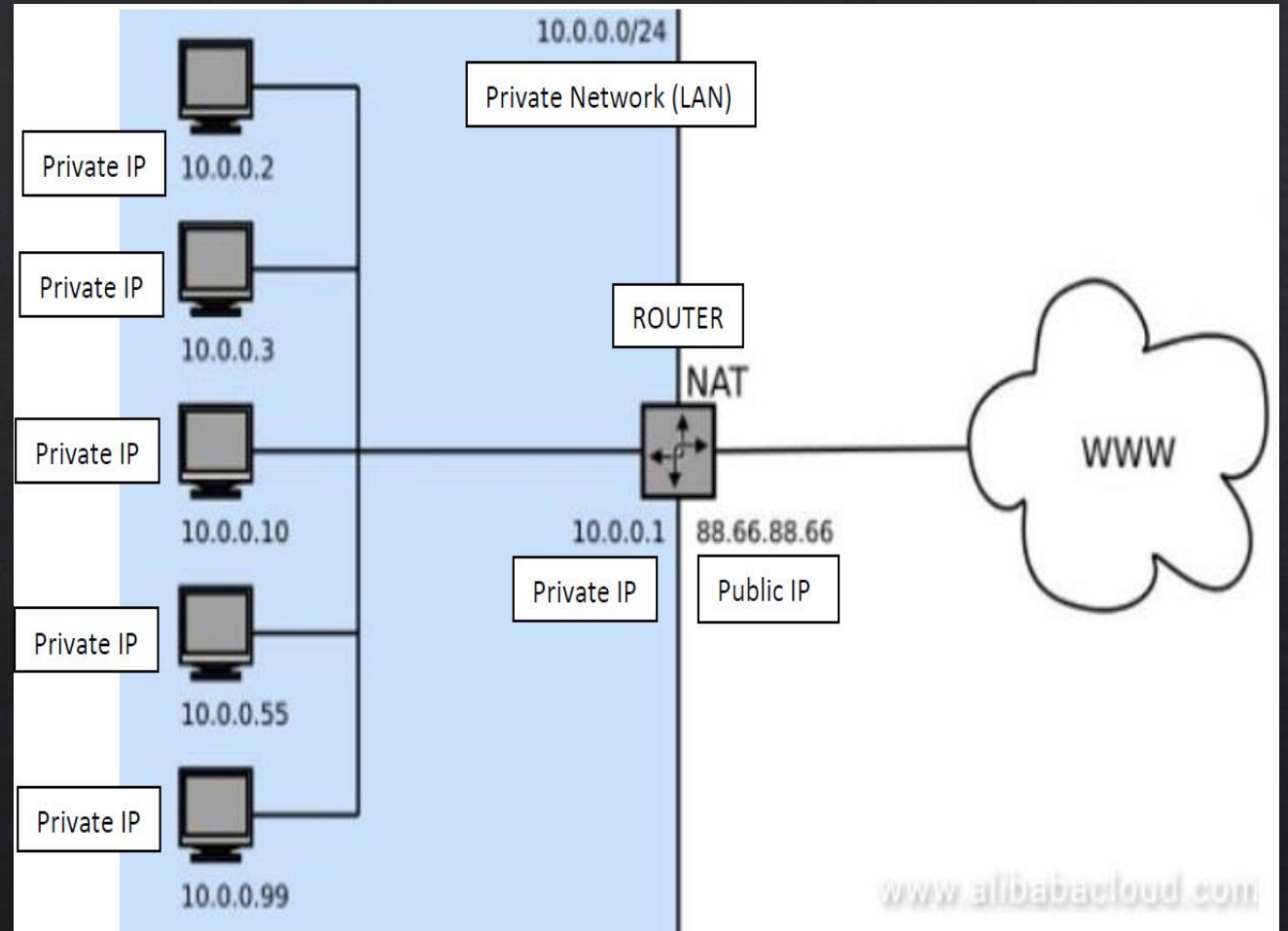
Network	Destination	Netmask	Gateway	Interface	Metric
	0.0.0.0	0.0.0.0	192.168.1.1	192.168.1.3	50

Routing Protocols

- ◇ Routing Protocols are responsible for determining:
 - ◇ Next hop.
 - ◇ Shortest path.
 - ◇ Network changes.
 - ◇ Link failures.
- ◇ Protocols types:
 - ◇ RIP: determine the shortest path and broadcast the routing table every 30 seconds.
 - ◇ OSPF: detect changes in the network topology and link failures.
 - ◇ BGP: most widely used protocol, can also determine the shortest path and if one route fail it changes to another route.

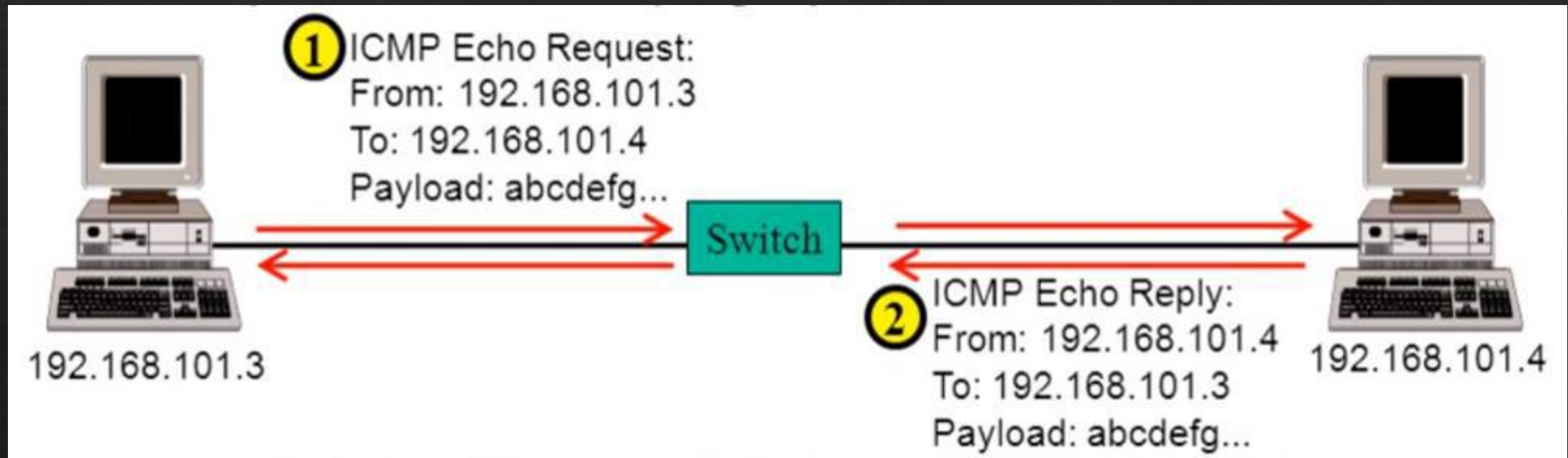
NAT

- ❖ Network Address Translation (NAT).
- ❖ It's a technique to translating one IP address to one or more IP addresses.
- ❖ All the home networks (LAN) is using the NAT protocol.



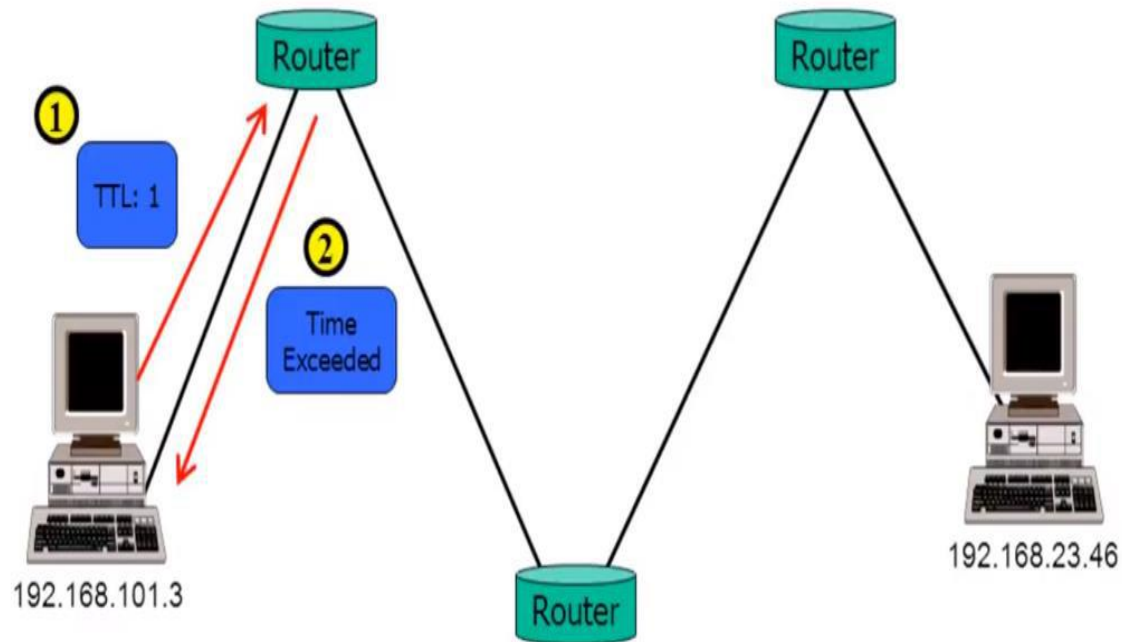
ICMP Protocol

- ◇ Test connectivity between two hosts.
 - ◇ Ping google.com

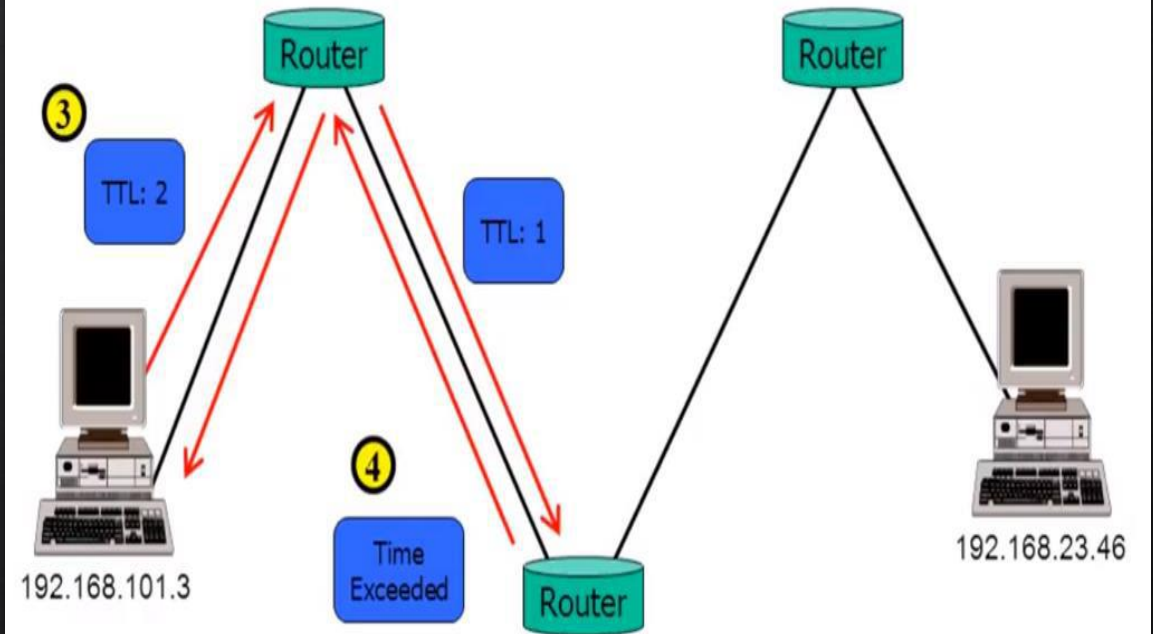


How Trace Route Work?

Traceroute Example (1)

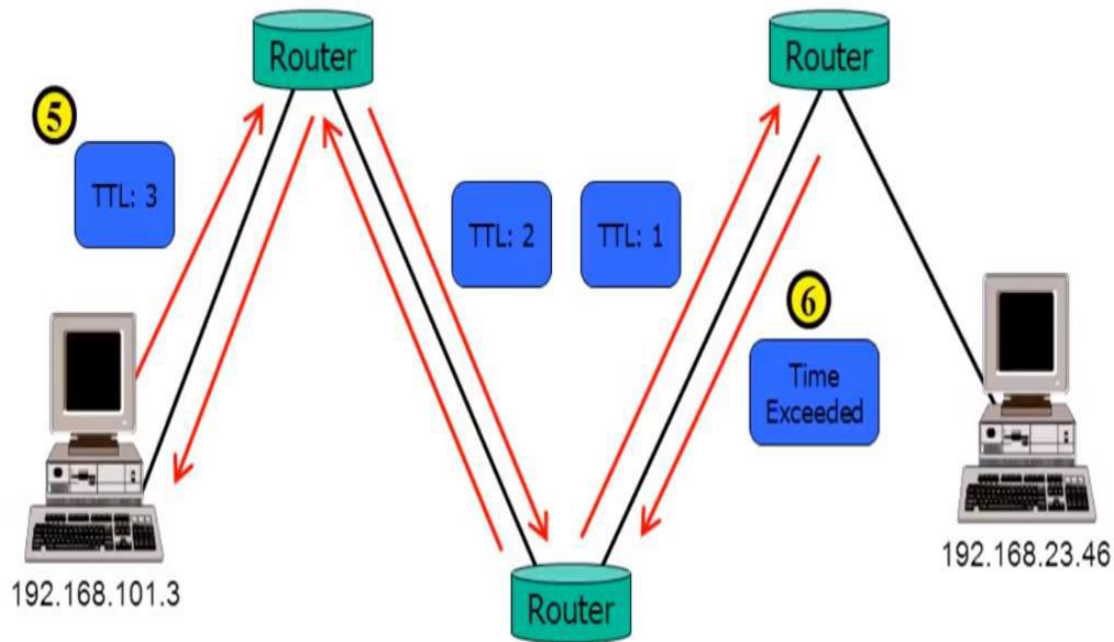


Traceroute Example (2)

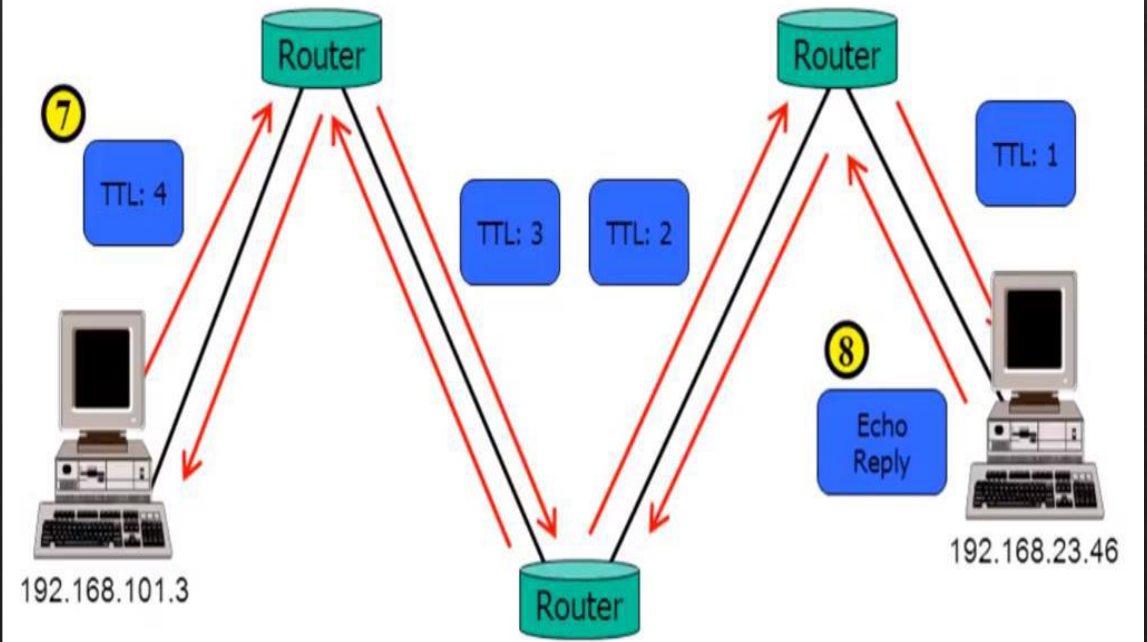


How Trace Route Work?

Traceroute Example (3)

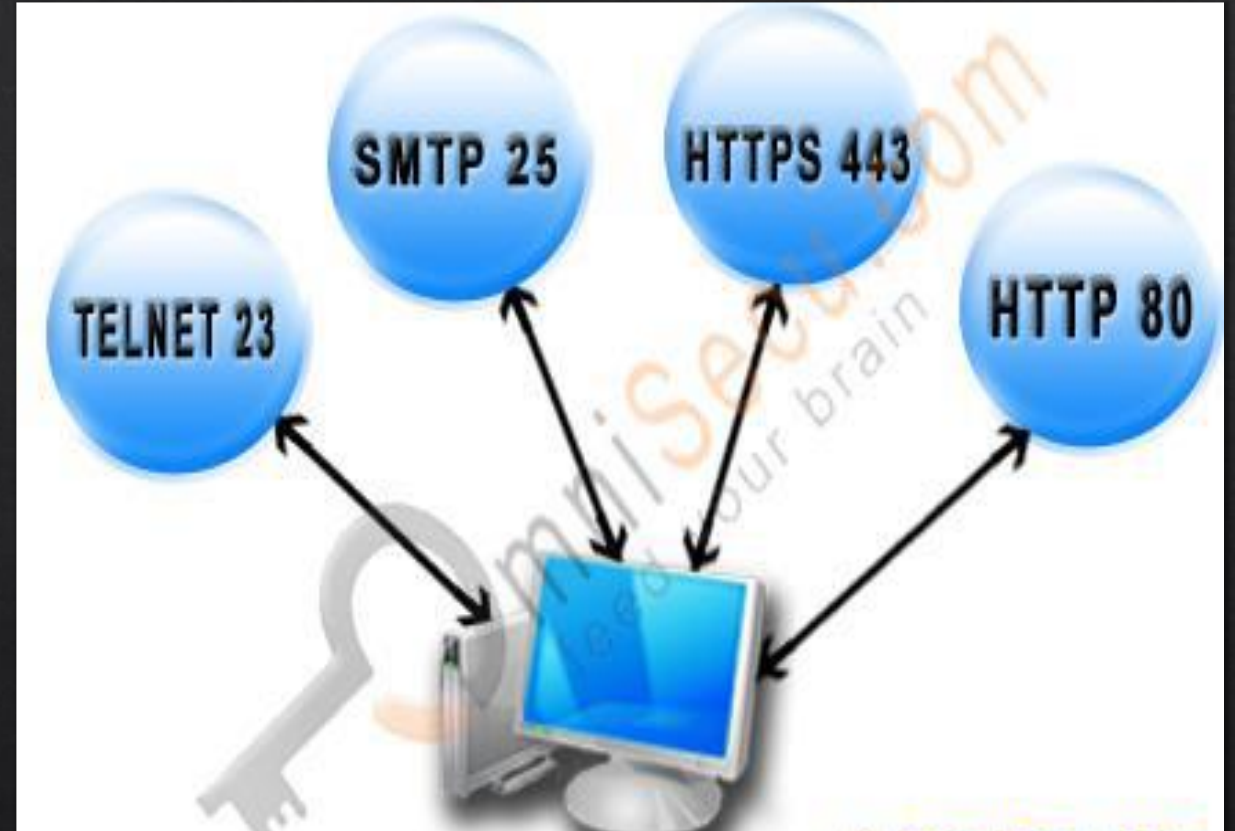


Traceroute Example (4)



Transport Layer (4)

- ◆ Ensure reliable data transfer between hosts.
- ◆ Determine the success transfers and the failed transfers and retransmit the failed ones and reorder the packets to form the original message.
- ◆ It also provide multiple ports on the same IP address, and it consists of mainly two protocols TCP, UDP.
- ◆ Ports are used to identify unique services on the same host.



Ports

- ◆ There are 65,536 ports on the TCP protocol and 65,536 ports on the UDP protocol.
- ◆ 1 – 1023:
 - ◆ Well known and the most used ports (HTTP, HTTPS, DNS, SMTP, SSH, FTP, TELNET).
- ◆ 1024 – 49151:
 - ◆ Available ports that can be used.
- ◆ 49152 – 65535:
- ◆ Can't be used because the operating system use this ports in the outgoing connections.

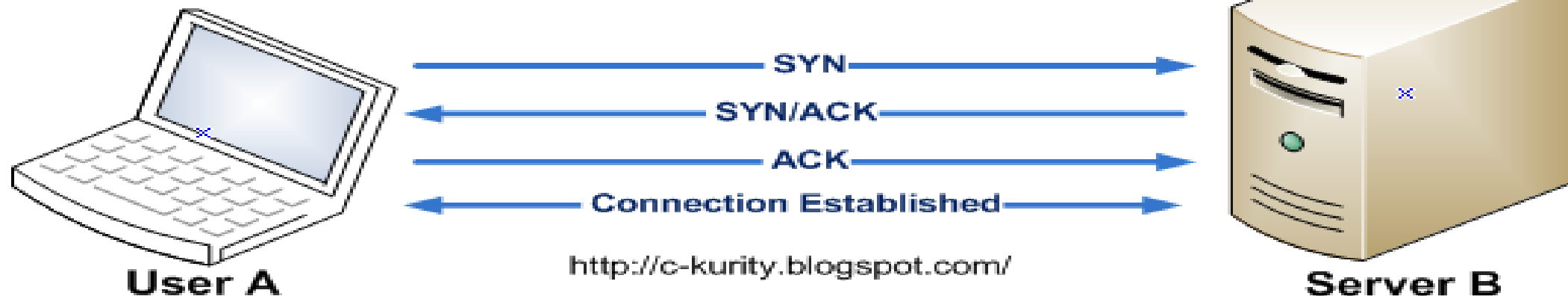
Important Ports to Know

- Here are some of the most important port numbers that you should be familiar with:
- 21: FTP (File Transfer Protocol)
- 22: SSH (Secure Shell)
- 23: Telnet
- 25: SMTP (Simple Mail Transfer Protocol)
- 53: DNS (Domain Name System)
- 80: HTTP (web traffic)
- 110: POP (Post Office Protocol)
- 135: MSRPC (Windows)
- 139: NetBIOS (Windows)
- 143: IMAP
- 443: HTTPS (SSL/TLS)
- 445: SMB/CIFS (Windows)
- 631: IPP/CUPS (Internet Printing Protocol)
- 3389: RDP (Terminal Services)
- 5800: VNC (Java viewer)
- 5900: VNC (Native client)

TCP Protocol

- ◆ TCP is the most used protocol.
- ◆ It performing a lot of functions to ensure the validation of data and a reliable connection.
 - ◆ Detect lost or failed data and retransmit it.
 - ◆ Filter if there is duplicate data found.
- ◆ TCP is design to ensure the accurate delivery not speed.
- ◆ TCP Flags:
 - ◆ It's a piece of information in the TCP header, add to every packet to help TCP protocol to ensure the accurate delivery.
- ◆ Flag Types:
 - ◆ Syn.
 - ◆ Ack.
 - ◆ Fin.
 - ◆ Push.
 - ◆ Reset.

TCP Three Way Handshake



No.	Time	Delta Time	Source	Destination	Protocol	Packet Length	Sequence number	HTTP Delta	Info
53	69.440126000	0.003658000	10.3.1.53	10.1.2.5	TCP	60	0	0	6713→2000 [SYN] Seq=0 Win=16000 Len=0 MSS=1456
54	69.586965000	0.146839000	10.1.2.5	10.3.1.53	TCP	60	0	0	2000→6713 [SYN, ACK] Seq=0 Ack=1 Win=5840 Len=0 MSS=1460
55	69.594753000	0.007788000	10.3.1.53	10.1.2.5	TCP	60	1	1	6713→2000 [ACK] Seq=1 Ack=1 Win=16000 Len=0

+	Frame 53: 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface 0
+	Ethernet II, Src: Cisco_6c:f4:4e (00:07:0e:6c:f4:4e), Dst: Cisco_09:56:42 (00:24:c4:09:56:42)
+	Internet Protocol Version 4, Src: 10.3.1.53 (10.3.1.53), Dst: 10.1.2.5 (10.1.2.5)
+	Transmission Control Protocol, Src Port: 6713 (6713), Dst Port: 2000 (2000), Seq: 0, Len: 0
	Source Port: 6713 (6713)
	Destination Port: 2000 (2000)
	[Stream index: 1]
	[TCP Segment Len: 0]
	Sequence number: 0 (relative sequence number)
+	Acknowledgment number: 86216704
	Header Length: 24 bytes
+	 0000 0000 0010 = Flags: 0x002 (SYN)

No.	Time	Delta Time	Source	Destination	Protocol	Packet Length	Sequence number	HTTP Delta	Info
53	69.440126000	0.003658000	10.3.1.53	10.1.2.5	TCP	60	0		6713→2000 [SYN] Seq=0 Win=16000 Len=0 MSS=1456
54	69.586965000	0.146839000	10.1.2.5	10.3.1.53	TCP	60	0		2000→6713 [SYN, ACK] Seq=0 Ack=1 Win=5840 Len=0 MSS=1460
55	69.594753000	0.007788000	10.3.1.53	10.1.2.5	TCP	60	1		6713→2000 [ACK] Seq=1 Ack=1 Win=16000 Len=0

Frame 54: 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface 0

Ethernet II, Src: Cisco_09:56:42 (00:24:c4:09:56:42), Dst: Cisco_6c:f4:4e (00:07:0e:6c:f4:4e)

Internet Protocol Version 4, Src: 10.1.2.5 (10.1.2.5), Dst: 10.3.1.53 (10.3.1.53)

Transmission Control Protocol, Src Port: 2000 (2000), Dst Port: 6713 (6713), Seq: 0, Ack: 1, Len: 0

Source Port: 2000 (2000)

Destination Port: 6713 (6713)

[Stream index: 1]

[TCP Segment Len: 0]

Sequence number: 0 (relative sequence number)

Acknowledgment number: 1 (relative ack number)

Header Length: 24 bytes

.... 0000 0001 0010 = Flags: 0x012 (SYN, ACK)

No.	Time	Delta Time	Source	Destination	Protocol	Packet Length	Sequence number	HTTP Delta	Info
53	69.440126000	0.003658000	10.3.1.53	10.1.2.5	TCP	60	0		6713→2000 [SYN] Seq=0 Win=16000 Len=0 MSS=1456
54	69.586965000	0.146839000	10.1.2.5	10.3.1.53	TCP	60	0		2000→6713 [SYN, ACK] Seq=0 Ack=1 Win=5840 Len=0 MSS=1460
55	69.594753000	0.007788000	10.3.1.53	10.1.2.5	TCP	60	1		6713→2000 [ACK] Seq=1 Ack=1 Win=16000 Len=0

Frame 55: 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface 0

Ethernet II, Src: Cisco_6c:f4:4e (00:07:0e:6c:f4:4e), Dst: Cisco_09:56:42 (00:24:c4:09:56:42)

Internet Protocol Version 4, Src: 10.3.1.53 (10.3.1.53), Dst: 10.1.2.5 (10.1.2.5)

Transmission Control Protocol, Src Port: 6713 (6713), Dst Port: 2000 (2000), Seq: 1, Ack: 1, Len: 0

Source Port: 6713 (6713)

Destination Port: 2000 (2000)

[Stream index: 1]

[TCP Segment Len: 0]

Sequence number: 1 (relative sequence number)

Acknowledgment number: 1 (relative ack number)

Header Length: 20 bytes

.... 0000 0001 0000 = Flags: 0x010 (ACK)

UDP Protocol

- ◇ UDP operate at the same level as TCP.
- ◇ UDP is connectionless and stateless:
 - ◇ No handshake.
 - ◇ No failure packet detection.
 - ◇ No retransmission.
- ◇ And because of this UDP is faster than TCP, and it used in cases where the accuracy not important like audio/video streaming where one packet lost don't effect the transmission.

Session (5) / Presentation (6)

- ◇ Session layer:
 - ◇ creates and terminates the unique connections between hosts.
- ◇ Presentation layer:
 - ◇ is responsible for encoding and decoding the message and show the message in the right format (Text, Photo, Video).

Application Layer (7)

- ◆ Application layer is the end user interface like web browser, mail client and so on.
- ◆ We will talk about two major application protocols: DNS, HTTP.
- ◆ DNS Protocol:
 - ◆ It's used to convert Hostname to IP address like: google.com => 172.217.21.78
 - ◆ Operate on UDP port 53.
- ◆ HTTP Protocol:
 - ◆ It's responsible for transferring the web pages and other files on the World Wide Web.
 - ◆ Operate on TCP port 80.

DNS Protocol

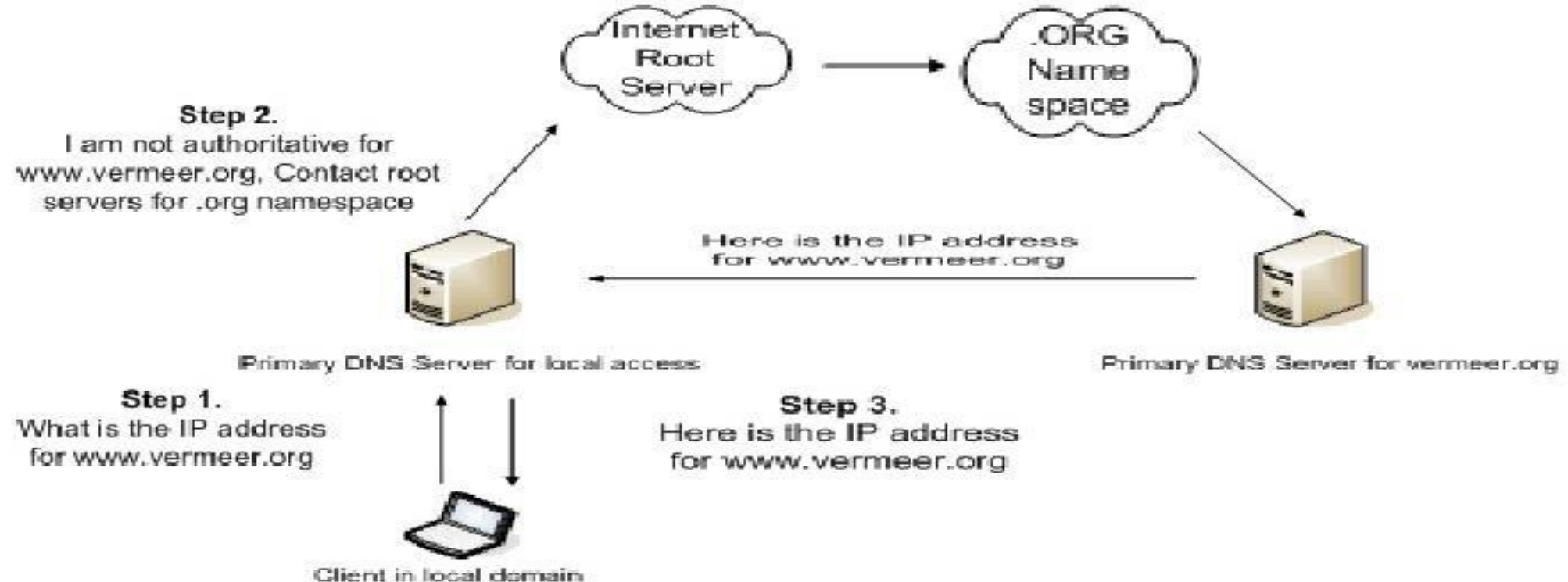
- ◆ Domain structure: (mail.google.com, login.yahoo.com)
 - ◆ Top level domain: .com, .net, .org, .edu, .gov
 - ◆ Second level domain: google, yahoo, facebook.
 - ◆ Sub-domains: mail, login.
- ◆ DNS record types:

Type	Description	Example
A	Show IP address.	nslookup -type=A google.com
MX	Show mail server.	nslookup -type=MX google.com
NS	Show DNS server.	nslookup -type=NS google.com

How DNS Work?

Example of a DNS query

DNS Name Resolution Request



HTTP Protocol

- ◆ HTTP is stateless so if you visit a login page and insert your username and password and now you have access on your account but you closed the page and revisiting it, then you will require to insert your username and password again, it don't remember you, it stateless.
- ◆ So now websites use cookies: it's a piece on information sent in the HTTP header to make the website remember you and don't ask your password every time you visit it.
- ◆ HTTP methods (GET, POST):
 - ◆ Both used to retrieve data from website.
 - ◆ But GET pass the variables in the URL and POST pass the variables in the HTTP header.
- ◆ HTTP Status Code:
 - ◆ 200: Success.
 - ◆ 300: Redirect.
 - ◆ 400: Not Found.

HTTP Protocol

Example HTTP Request

GET /aboutus.php HTTP/1.1

Host: target.securitytreasurehunt.com

User-Agent: Mozilla/5.0 (Macintosh; Intel Mac OS X 10.6; rv:7.0) Gecko/20100101 Firefox/7.0

DNT: 1

Connection: keep-alive

Example HTTP Response

HTTP/1.1 200 OK

Date: Fri, 30 Sep 2011 15:32:11 GMT

Server: Apache

Content-Length: 3367

Connection: Keep-Alive

Content-Type: text/html

```
<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0  
Strict//EN"  
"http://www.w3.org/TR/xhtml1/DTD/xhtml1-  
strict.dtd">
```